Biometric Beats: Elevating Crowd Experiences with Adaptive Music

Grace Liu (gl2910)
Darien Moment (dem2187)
Matthew Rostkowski (mcr2225)

1 INTRODUCTION

Music plays a central role in shaping social experiences, whether at parties, concerts, or community events. However, that introduces a myriad of obstacles, including anxiety, when tasked with trying to find suitable music for a group of people in a specific context or when people are in situations with limited song-selection power. With the increasing prevalence of wearables such as Apple watches and fitness trackers, human-computer interaction (HCI) has the opportunity to leverage these technologies and real-time biometric data to better support shared experiences. The present project investigates how biometric data, specifically heart rate, can be used to dynamically optimize the musical experience in group settings.

We propose an adaptive music system that integrates real-time biometric feedback from wearable devices to adjust music selections based on collective audience engagement. The goal is to create a context-aware musical experience that passively responds to crowd dynamics to enhance enjoyment without the need for active user input. This approach blends growing trends in biometric HCI, adaptive systems, and music technologies. We introduce a smart music control interface with biometric inputs that reduces the stress of individual control — "aux anxiety" — while fostering a more emotionally attuned musical environment.

Together, this research has three main contributions to the HCI community:

- **An interactive system (artifact):** The system integrates Apple HealthKit to collect real-time heart rate data from Apple watches, enabling a context-aware, adaptive DJ Experience.
- **A new interaction model:** Our passive, group-level interaction model uses aggregated heart rate data to adjust music selections in real time, reducing decision fatigue and easing "aux anxiety."
- **Expansion of biometric feedback HCI research:** This project builds on biometric feedback HCI research by addressing a gap in real-time, group-level music adaptation. While prior work has explored mood-based and individual biometric music systems, few have investigated collective biometrics in social settings.

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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

2 RELATED WORKS

Our work builds on previous research in mood-based music systems, biometric sensing in interactive applications, and collective user experiences. While each area has seen meaningful contributions, the intersection of real-time, group-level biometric feedback and adaptive music systems remains underexplored, particularly in socially shared contexts.

2.1 Mood-based and Adaptive Music Systems

Research on adaptive music systems has explored personalization based on emotional states. Moodplay is a mood-driven music discovery platform that enables users to explore and control music through arousal and valence emotional dimensions through crowdsourced mood information and static metadata, supporting individual and collective control through mood voting and trajectory drawing [1]. However, Moodplay and similar systems [2] rely on static metadata or manual input, limiting their responsiveness to users' real-time emotional states. Other systems have experimented with mood detection via facial expression recognition [3], but this targets individual users rather than groups.

2.2 Biometric Feedback in HCI

Biometric data has become increasingly common in HCI, offering a rich, continuous feedback stream for adaptive environments. Heart rate has been used to dynamically control a game's skiing speed and shooting accuracy [4]; galvanic skin response has enabled binary communication for locked-in patients [5]; and brain waves have been translated into collaborative musical compositions [6]. Sparrow et al. [7] explored the best methods for collecting audiences' biometric data and its ethical and practical challenges during immersive performances.

Commercial platforms have also tapped into this biometric potential. Spotify's now-retired running feature detected a user's running tempo to adjust music playback to match that pace [8], but it was designed for individual rhythm rather than emotional or group dynamics.

2.3 Collective Interaction and Shared Experiences

Recent work has investigated how group-level biometric data can support co-located or remote collective experiences. Moge et al. [9] detail social biofeedback systems where users share real-time physiological data (e.g., heart rate, EEG) to enhance collaboration and emotional synchronization. Similarly, real-time, multi-user biometrics have also been used to passively guide generative AI [10].

Our work extends this emerging work by leveraging passive, aggregated heart rate data to drive real-time music adaptation. By

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Darien Moment (dem2187)
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using biometric signals as a proxy for crowd engagement, we create a socially synchronized music experience without requiring active user input. In doing so, we address a key gap in HCI: bringing together real-time, group-level biometric feedback and adaptive music to support shared emotional experiences.

3 SYSTEM ARCHITECTURE

3.1 Heart Rate-Based System

The Heart Rate based system shares real-time heart rate data from Apple Watches using Apple's HealthKit framework (Fig. 1). Reported heart rates are stored in a Firebase Realtime Database, and retrieved by a Python server that aggregates data from all connected watches. During each song, the server calculates the average heart rate across all participants. Once the song is over, it calculates the overall average for the duration of the song, and compares it to the previous song's average. If the current average is greater than or equal to the previous average, a prompt is sent to GPT-3.5 Turbo for a similar song; otherwise, a dissimilar song is requested. The suggested song is then sent to the researchers to play next.



Figure 1: Apple Watch interface used to collect real-time heart rate data, which is aggregated and used to guide adaptive music transitions based on group engagement.

3.2 Voting-Based System

The voting based system follows the same workflow as the heart rate system, yet sends votes instead of heart rates (Fig. 2). Each participant uses a simple voting interface to upvote or downvote the current song. Votes are stored in the Firebase Realtime Database, and retrieved by a Python server, which determines the majority vote. If the majority are upvotes, a prompt for a similar song is sent to GPT-3.5 Turbo; otherwise, a dissimilar song is requested. The suggested song is then sent to the researchers to play next.

3.3 Monitoring-Based System

The monitoring-based system resembles a traditional DJ setup. We observe participant movement during each song. If increased movement is observed, a prompt for a similar song is sent to GPT-3.5 Turbo. If decreased movement is observed, a dissimilar song is requested. The suggested song is then sent to the researcher to play next.



Figure 2: Apple Watch interface used for the voting-based system, allowing participants to upvote or downvote songs in real time during playback.

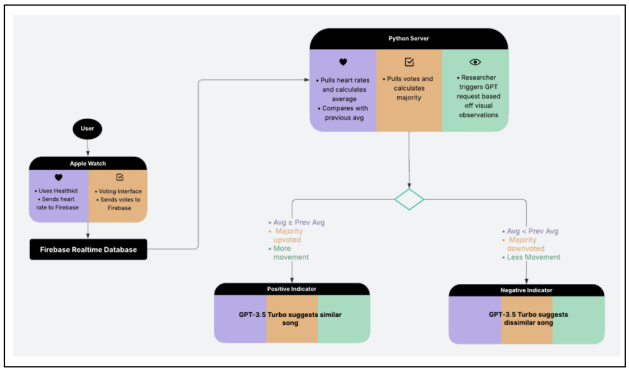


Figure 3: System architecture for Biometric Beats. The pipeline integrates real-time data from Apple Watches—either heart rate, voting input, or researcher observation—into a shared Firebase database. A Python server processes the input to determine whether to request a similar or dissimilar song from GPT-3.5 Turbo based on crowd engagement signals.

4 METHODS

4.1 Study Design

We conducted a within-subjects user study to evaluate the three different music control systems on group engagement and user experience in a social context. Three participants (ages 21–22), recruited from Columbia University, took part in the study as a single group. The study was designed to simulate a casual group music setting, with participants encouraged to dance, react, and interact naturally throughout the session.

4.2 Procedure

Starting with the heart rate-based system, participants first completed a one-minute resting heart rate measurement to establish a baseline. They then listened and danced for the duration of 12 songs. After the listening session, we conducted semi-structured interviews to collect qualitative feedback on participants' experiences. This procedure was repeated for the voting-based and monitoring-based systems, excluding the initial heart rate collection.

5 RESULTS

The effectiveness of the different adaptive-music methods was quantified primarily by the percentage of sessions in which a positive indicator was reached. This is because the positive-indicator percentage provides a common, model-free metric that treats each song switch as a binary success or failure, allowing direct, apples-to-apples comparison of how reliably each method elicits favorable listener responses across trials.

5.1 Heart Rate–Based System

Among the three adaptive–music methods, heart–rate switching produced the highest engagement: a positive indicator was recorded in 77.8% of sessions, meaning listeners' physiological arousal increased after a track change (Fig. 8). Post–session surveys echoed this outcome, giving the system a mean usability score of 8.8/10. Participants especially appreciated its invisible operation, one noting, "I like that I knew I was influencing the music without actually consciously thinking about it, which made the experience feel effortless." A minority expressed concern that normal pauses could be misinterpreted, as reflected in another comment: "There were times when I enjoyed the song but maybe I just needed to rest, so I became a little bit 'paranoid' that I was causing the system to think I wanted a shift in the music when I didn't."

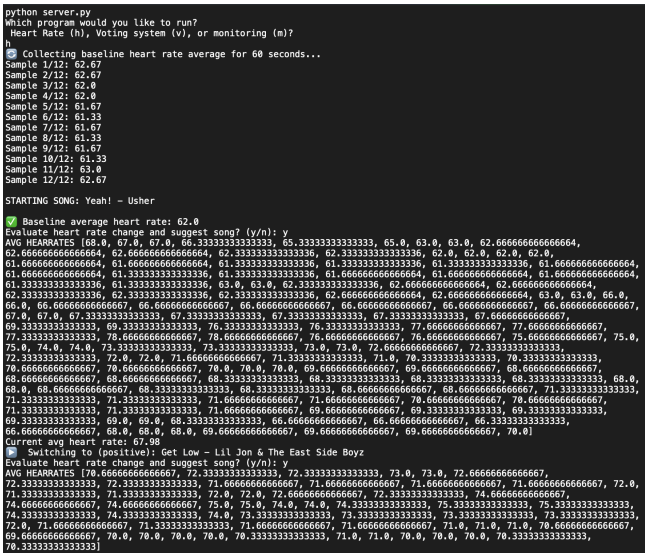


Figure 4: Heart-Rate System sample logs

5.2 Voting–Based System

The voting interface ranked second, with a 70.4% positive indicator rate and a usability score of 8.3/10. Its binary up–vote/down–vote design was praised for clarity, yet multiple users wished for finer granularity: "There were times when the song was solid, but it was not my most favorite, and that made it really hard to decide whether to up-vote or down-vote, so I sometimes withheld feedback."


```
"Votes": {
  "watch1": {
    "state": "upvoted",
    "timestamp": "2025-05-06T20:52:22Z",
    "value": 1
  },
```


Figure 5: Voting System sample logs

5.3 DJ/Vision–Based System

Traditional DJ/vision cues proved least effective, achieving a 59.5% positive rate. Listeners cited delays between their visible reactions and the DJ's response, yielding a lower seamlessness rating of 6.9/10. Many resorted to exaggerated gestures to be noticed, underscoring the limitations of vision alone in crowded settings.

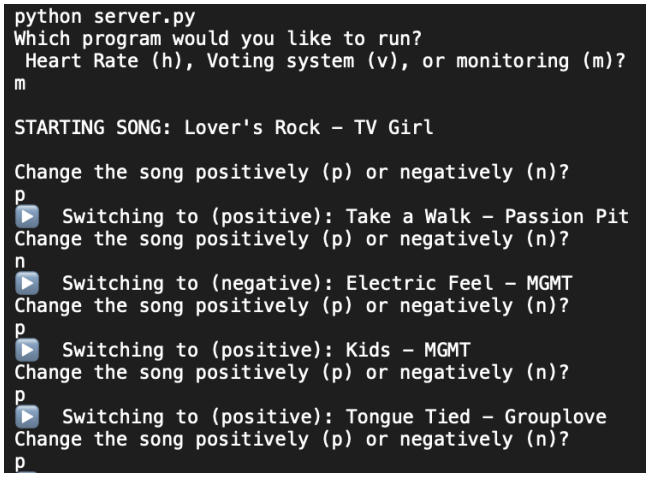


Figure 6: DJ/Vision–Based System sample logs

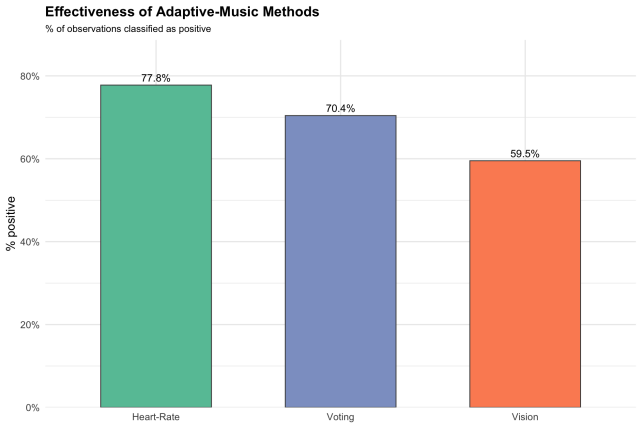


Figure 7: Positive transition rates for each adaptive–music method: heart rate (77.8%), voting (70.4%), and DJ/vision (59.5%).

5.4 95% Confidence Intervals

Comparing the 95% confidence intervals for the three methods' positive-indicator percentages reveals important differences in statistical precision. For the heart-rate system the interval spans 45.3-93.7%, for voting it narrows to 59.0–79.8%, and for DJ/vision it sits at 48.8–69.4%. The much wider band for heart rate is unsurprising: participants frequently paused to rest, causing drops in BPM, and the sample contained only a sample size of transitions between songs, so each success or failure shifted the estimate substantially. By contrast, voting produced a tighter interval because every data point was an intentional, binary decisions by participants, thereby reducing random noise. DJ/vision falls in between, yet subjective delays in the DJ's response still introduce variability that keeps the interval broader than that of voting.

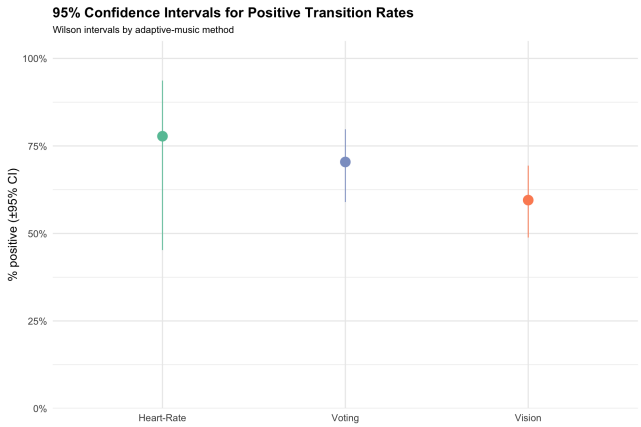


Figure 8: 95% confidence intervals for positive response rates across adaptive-music methods.

6 DISCUSSION

These findings confirm that passive biometric feedback can outperform explicit voting and traditional DJ intuition in sustaining collective engagement. Heart–rate adaptation reduced decision fatigue and heightened immersion, whereas binary voting demanded continuous judgments from listeners, and DJ/vision suffered from reaction lag.

To smooth abrupt genre shifts triggered by single negative cues, future iterations should incorporate short–memory forecasting. An ARIMA controller that weights responses from the last few tracks could distinguish temporary lows from genuine disengagement, thereby maintaining flow without chasing a perpetually rising curve. Such temporal modeling would respect natural cycles of activity, rest, and re-engagement while preserving the effortless influence that listeners found so appealing.

7 LIMITATIONS AND FUTURE WORK

This paper introduced an exploration into adaptive music systems driven by biometric feedback. Although our system demonstrates the feasibility of real-time song selection based on physiological data, there are several limitations. First, we were limited to three Apple watches for testing, which limits the generalizability of our results. Second, we relied on GPT-3.5 Turbo to suggest songs; however, the model's recommendations may not always align with participants' preferences. Finally, all tests were conducted in dorm lounges and suites, environments that differ greatly from the social and dynamic conditions of public venues.

These limitations open up several directions for future work. We plan to scale the system to support larger groups by including more Apple watches or exploring other wearable devices. To ensure that more relevant song suggestions are generated, we plan to incorporate contextual information such as genre preferences, participant moods, and environmental factors. Moreover, testing the system in more dynamic, high-energy settings (e.g., nightclubs, concerts, or festivals) will provide insight into how well our system performs.

In addition to limitations, our system presents important ethical considerations. Key ethical concerns include privacy, autonomy, trust, and the potential for biometric data to reveal sensitive personal information (e.g., health status, emotions, or even sexual orientation) or be misused (e.g., for targeted advertising) [7]. As our system is developed and implemented in the real world, it will be crucial to ensure informed consent, transparent data handling procedures, and strong protections against misuse.

8 CONCLUSION

This work presents Biometric Beats, a novel music system that harnesses real-time, group-level biometric data for adaptive, shared musical experiences. Our system uses aggregated heart rates from Apple watches to determine crowd engagement and guide music transitions without the burden of manual control. Our user study found that participants felt more engaged and less stressed with the heart rate–driven system compared to manual voting or DJ-controlled setups. These findings suggest that biometric feedback can support more seamless, enjoyable group experiences by responding to collective mood in real time. While limited in scale, this work introduces a novel interaction model and contributes to ongoing research in adaptive systems and social HCI. Future work will focus on expanding deployment to more watches and larger venues, building away from GPT-3.5 dependency by adding contextual data (e.g., genre preferences, mood) or creating our own recommendation system, and addressing ethical concerns around biometric data use. Ultimately, Biometric Beats demonstrates the vast potential of real-time physiological data to elevate shared musical experiences in passive ways — bringing us closer to technologies that adapt to not just what we do, but how we feel.

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